

UNIVERSITY OF HAMBURG

MASTER THESIS

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# Masterthesis\_doc

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*Author:*

Lona Frießner

*Supervisor:*

Prof. Dr. Simon Grund

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*in the*

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UNIVERSITY OF HAMBURG

## *Abstract*

Institute of Psychology

Department of Psychology with focus on Quantitative Methods

Master of Science

**Masterthesis\_\_doc**

by Lona Frießner

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## Chapter 1

# Introduction



## Chapter 2

# Theory



## Chapter 3

# Multilevel models

### Example

Imagine that you are a researcher interested in the musical self-concept of young adults. You want to examine whether there is a connection to weekly duration of practice on their instrument.

So you gather a sample. You manage to find 15 youth ensembles that contribute to your study. For simplicity, let's consider that there are 10 players in each ensemble. Each player gets a questionnaire with items about their musical self-concept and their weekly practice duration.

### 3.1 The idea behind multilevel models

Populations like this, where units of interest (e.g., music players, students, employees) are grouped into higher-level units (e.g., ensembles, classes, firms), are common in the human social life and therefore in psychological studies.

The resulting data is called *nested* or *clustered* (Snijders & Bosker, 2012, p. 7).

From a sampling perspective, a so-called *multistage sampling process* (sample groups first, then individuals inside these groups<sup>1</sup>) is not independent: After selecting a group, you will try to sample all of the members of this group, making it more likely to select one person in this group over others in non-selected groups (Snijders & Bosker, 2012, p. 7). Also, individuals inside the same group typically share contextual influences (e.g., the same ensemble teacher, the same class room) and thus will give more similar responses in surveys and behavior. As a consequence, the basic assumption of independence in traditional linear modeling is violated, because error terms will be correlated (Raudenbush & Bryk, 2010, p. xx).

One well-suited and popular possibility to account for this fact is the use of *multilevel models* (also called *hierarchical linear models* or *mixed models*).

A multilevel model (MLM) extends regular linear regression by adding *random effects*

### 3.2 The random intercept model with context effects

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<sup>1</sup>For simplicity, this thesis does not address three (or higher)-level designs.





## Chapter 4

# Small sample sizes in multilevel models

Auch Bayesian, Raudenbush S. 399



## Chapter 5

# Missing data in multilevel models

### 5.1 Methods for handling missing data

#### 5.1.1 Listwise Deletion

#### 5.1.2 Full Information Maximum Likelihood

#### 5.1.3 Bayesian Model Estimation

#### 5.1.4 Multiple Imputation



## Chapter 6

# Methods

A simulation study was conducted in order to compare the performance of methods for missing data handling in small-sample multilevel random intercept models.

The idea behind a simulation study is to carry out a computer experiment (Morris et al., 2019), where data with known characteristics is simulated and subsequently analyzed with the methods of interest. The analysis results can be compared to the truth, allowing for assessment of the performance of the methods (Siepe et al., 2024).

Planning of the simulation study followed the ADEMP structure proposed by Siepe et al. (2024) and will be reported based on their guidelines.

### 6.1 Data generation mechanism

#### 6.1.1 Data-generating model

The data-generating model is a two-level random intercept model:

$$Y_{ij} = \gamma_{10}(X_{ij} - \bar{X}_{\cdot j}) + \gamma_{01}\bar{X}_{\cdot j} + \gamma_{02}W_j + u_{0j} + e_{ij} \quad (6.1)$$

$Y_{ij}$ ,  $X_{ij}$  and  $W_j$  are parameterized to have a mean of zero and a variance of 1 on the population level.

First, between- and within-components of  $X_{ij}$  were simulated, independent of each other and with variances chosen to achieve a prespecified intraclass coefficient (see Section 6.1.3):

$$\begin{aligned} X_{ij} &= u_{Xj} + e_{Xij}, \\ u_{Xj} &\sim N(0, \tau_x^2), \quad e_{Xij} \sim N(0, \sigma_x^2) \end{aligned}$$

with  $\tau_x^2 = ICC_x$  and  $\sigma_x^2 = 1 - ICC_x$ , respectively.

$X_{ij}$  was then group-mean-centered.

The group-level covariate  $W_j$  was constructed with a linear regression equation to have a prespecified correlation with  $\bar{X}_{\cdot j}$  ( $\rho_{\bar{X}W}$ ):

$$W_j = a\bar{X}_{\cdot j} + e_{Wj}, \quad e_{Wj} \sim N(0, \sigma_w^2)$$

The residual variance  $\sigma_w^2$  and the regression coefficient  $a$  are fully determined by the restriction that  $Var(W_j) = 1$ :

$$\begin{aligned} \sigma_w^2 &= 1 - \rho_{\bar{X}W}^2, \\ a &= \frac{\rho_{\bar{X}W}}{\sqrt{\tau_x^2 + \frac{\sigma_x^2}{n}}} \end{aligned}$$

TABLE 6.1: Simulation conditions for the Data-Generating Model and the Generation of Missing Values

| <b>Data conditions</b>              |              |
|-------------------------------------|--------------|
| $N_1$                               | 10           |
| $N_2$                               | 15 / 30 / 60 |
| ICC of $X$ and $Y$                  | .1 / .3      |
| Correlation $\bar{X}W$              | .3           |
| <b>Model parameters</b>             |              |
| Effect of $(X_{ij} - \bar{X}_{.j})$ | .2           |
| Effect of $\bar{X}_{.j}$            | 0 / .4       |
| Effect of $W_j$                     | 0            |
| <b>Missing values</b>               |              |
| Pattern                             | $X \sim W$   |
| Mechanism                           | MCAR / MAR   |
| Proportion                          | .3           |

Finally, the random effects on the group and the individual level were generated as

$$u_{0j} \sim N(0, \psi^2), \quad e_{ij} \sim N(0, \sigma^2)$$

and independent of each other.

Individual and group residual variances of  $Y_{ij}$  are determined by the constraints of a prespecified residual ICC of  $Y_{ij}$ ,  $Var(Y_{ij}) = 1$  and  $Var(X_{ij}) = 1$ <sup>1</sup>:

$$\psi^2 = ICC_{Y|X,W} \cdot (1 - \gamma_{10}^2 (\sigma_X^2 - \frac{\sigma_X^2}{n_j}) + \gamma_{01}^2 (\tau_x^2 + \frac{\sigma_X^2}{n_j}))$$

$$\sigma^2 = (1 - ICC_{Y|X,W}) \cdot (1 - \gamma_{10}^2 (\sigma_X^2 - \frac{\sigma_X^2}{n_j}) + \gamma_{01}^2 (\tau_x^2 + \frac{\sigma_X^2}{n_j}))$$

The full derivation of these equations is provided in [?@sec-appA](#).

### 6.1.2 Missing data generation

After simulation of the complete dataset, values of the  $X$ -variable have been deleted to simulate missingness. Following an approach by Grund (2023, unpublished HTML-Document), a ‘missingness tendency’  $r$  was simulated for each  $X$ -value according to this equation:

$$r_{ij} = \alpha + \beta W_j + e_{ij}, \quad e_{ij} \sim N(0, 1 - \beta^2)$$

$r_{ij}$  is a normally distributed, continuous variable. When  $r_{ij} > 0$ , the  $ij$ -person’s  $X$ -values was deleted.

The intercept  $\alpha$  controls the proportion of missingness in  $X$ , whereas the slope  $\beta$  determines the strength of the missing-at-random mechanism. When  $\beta = 0$ , missingness is completely at random (MCAR). With increasing  $\beta$ , the missingness mechanism becomes more dependent on the  $W$ -values; with  $\beta = 1$ , missingness would be completely deterministic.

### 6.1.3 Factors of the data-generating mechanism

A summary of simulation conditions is provided in Table 6.1.

$N_2$  was varied ( $N_2 = 15, 30, 60$ ), whereas  $N_1$  was held constant ( $N_1 = 10$ ). The effect of  $\bar{X}_{.j}$  was varied ( $\gamma_{01} = 0, .40$ ) in order to compare the methods’ behaviour with

<sup>1</sup>as well as setting  $\gamma_{02}$  to be zero across all conditions, see Section 6.1.3

a null effect and a substantial effect. The effect of  $(X_{ij} - \bar{X}_{.j})$  was held constant ( $\gamma_{10} = .20$ ) as well as the effect of  $W_j$  ( $\gamma_{02} = 0$ ). With these settings, there are conditions with either a negative and a positive context effect of  $X$  in the population model. The ICC of  $X$  and the residual ICC of  $Y$  were varied ( $\rho_X = \rho_{Y|X} = .10, .30$ ). Missing data were generated on  $X$  with a varying missingness mechanism (MCAR or MAR with a strength of  $\beta = .30$ ), with a constant missingness rate of 30%. The factors were combined fully factorial, creating  $3 \times 2 \times 2 \times 2 = 24$  conditions.

All in all, these settings reflect a number of the typical applications in cross-sectional and longitudinal applied research (e.g., smaller and larger intraclass correlation coefficients (ICC) and level-two sample sizes). They also create sufficient methodological variation to evaluate the properties of the methods of interest (e.g., null versus substantial effects).

Missing data mechanism is set to either MCAR or MAR. For MAR, the strength of relationship between  $W$  and missing of  $X$  is set to 0.4, which corresponds to  $0.4^2 = 16\%$  explanation of variance in missingness through  $W$ .

#### **6.1.4 Methods of missing data handling**

##### **6.1.5 Estimands**

##### **6.1.6 Performance measures**

##### **6.1.7 Execution of simulation**





## Chapter 7

# Results

7.1 Bias of point estimation

7.2 Coverage of confidence/credible intervals

7.3 Standard deviation of point estimates



## Chapter 8

# Discussion



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